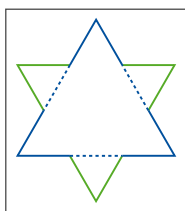
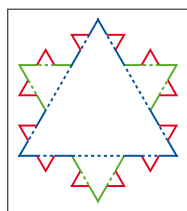


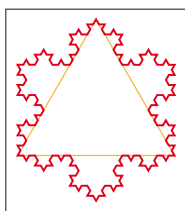
1



2



3



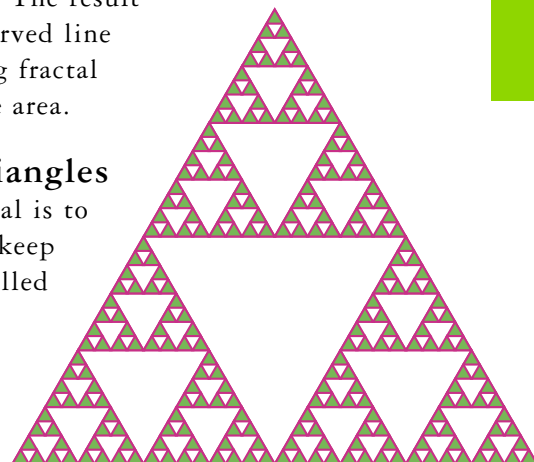
4

Koch's snowflake

Draw an equilateral triangle, then draw small equilateral triangles a third as wide on its sides. Do the same again, and carry on doing this forever. The result is a fractal called "Koch's snowflake" – a curved line made purely of corners. This mind-boggling fractal has an infinitely long perimeter but a finite area.

Triangles in triangles in triangles

Another simple way to make a fractal is to draw triangles inside triangles and keep doing this forever. This fractal is called "Sierpinski's gasket". Amazingly, if you colour in multiples of 2 or 3 in Pascal's triangle, you'll see a very similar pattern take shape.



Crinkly coastlines

How long is the coastline of North America? It's impossible to give a simple answer to this question because a coast is a fractal line. If you measured it on an atlas, you'd get one answer. If you used a more detailed map, you'd discover more wrinkles and would get a bigger answer. And if you drove or walked along the coast, you'd get a bigger answer still. The rate by which these measures increase as you zoom in is called the *fractal dimension*.

Ice ferns

Some of the most common natural fractals form when something branches over and over again, like a tree trunk splitting into boughs, branches, and twigs. We call this pattern "dendritic". Frost ferns, rivers and tributaries, and the veins in your body all make dendritic fractals.